**6** **(a)** Photons are produced whenever a **<u>charged particle</u>** such as electrons **<u>decelerates</u>**. **B1**

The electrons produced from the cathode will strike the metal target and decelerates. Since the electrons has a **<u>distribution of decelerations</u>**, the X-ray photons produced also has a distribution of wavelengths. Hence a continuous spectrum is produced. **B1**

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|---|---|
| **MC:** | *Generally very poorly done. Many candidates quoted either the explanation for characteristic emissions or phenomena completely unrelated to X-ray production such as line spectra, discrete energy levels, radioactive decay or photoelectric effect.*<br><br>*Amongst those that managed to produce some semblance of the above answer, most did not state explicitly the mechanism in which photons can be produced using charged particles. However, most were able to identify that the electrons will decelerate to different extents (or lose different amounts of KE) which gives rise to the continuous background.* |

**(b)** **(i)**

$$\frac{hc}{\lambda_{\text{min}}} = eV$$

$$\lambda_{\text{min}} = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{1.6\times10^{-19}\times10000}$$ **C1**

$$=1.2431\times10^{-10}$$

$$=1.24\times10^{-10}\text{m}$$ **A1**

| | |
|---|---|
| **MC:** | *Generally very poorly done. Most candidates could not link the minimum wavelength to the accelerating potential but instead attempted to use energy differences in the energy level diagram to calculate their answers.* |

**(ii)** For $K_\alpha$ line,

$$|E_2 - E_1| = \frac{hc}{\lambda_\alpha}$$

$$\lambda_\alpha = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{(-952+8980)\times1.6\times10^{-19}}$$ **C1**

$$=1.5485\times10^{-10}\text{ m}$$

$$=1.55\times10^{-10}\text{ m}$$ **A1**

For $K_\beta$ line,

$$\left|E_3 - E_1\right| = \frac{hc}{\lambda_\beta}$$

$$\lambda_\beta = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{(-75.0+8980)\times1.6\times10^{-19}}$$

$$=1.39599\times10^{-10}\ \text{m}$$

$$=1.40\times10^{-10}\ \text{m}$$ **A1**

**Note:** C1 awarded for any correct substitution for either $K_\alpha$ or $K_\beta$.

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|---|---|
| **MC:** | *Generally very poorly done. Most students were not aware of the energy level transitions that produces the $K_\alpha$ and $K_\beta$ lines. Despite it being labelled in the diagram to further aid students, many ended up with a $\lambda_\alpha < \lambda_\beta$.* |

**(c)** By Heisenberg's Uncertainty principle,

$$\Delta p \Delta x \geq h$$

$$m \Delta v \Delta x \geq h$$

$$\Delta v \geq \frac{h}{m\Delta x} = \frac{6.63\times10^{-34}}{9.11\times10^{-31}\times10^{-15}} = 7.3\times10^{11}\ \text{m s}^{-1}$$ **C1**

Since the required velocity of electrons $\Delta v$ is greater than the speed of light of $3.0\times10^{8}$ m s$^{-1}$, the electrons does not exist inside the nucleus. **A1**

| | |
|---|---|
| **MC:** | *Generally very poorly done. Quite a significant number did not attempt the question. Most candidates who attempted the question were also unable to make connections to the speed of light.*<br><br>***<u>Common errors:</u>***<br>• *Using the de Broglie wavelength equation to do the calculations. (either to compute p or λ to be used in the Heisenberg's Uncertainty relation)*<br>• *Stating that the electron has a momentum of $\sqrt{2mE} = \sqrt{2mqV}$ where V is the accelerating potential in the earlier part. (This is incorrect as the V is an externally applied potential difference. The electron will not exist in the nucleus even in the absence of this potential difference.)* |

7 **(a)** Neutron **A1**